

Installing ROS 2 Humble and Gazebo Fortress on Windows using WSL2

Overview

This guide walks you through installing **ROS 2 Humble Hawksbill** and **Gazebo Fortress** on **Windows 10/11** using the **Windows Subsystem for Linux 2 (WSL2)**.

You'll end up with a full ROS 2 + Gazebo simulation environment, ready for running navigation, visualization, and control experiments — all inside Windows.

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0. Verify and Install WSL2

What is WSL2?

WSL2 lets you run a full Linux environment (Ubuntu) directly inside Windows — no dual booting or virtual machines required. It is necessary for running ROS 2 and Gazebo on Windows.

Step 1: Check if WSL is installed

Open **PowerShell as Administrator** and run:

```
#run the first two commands in the terminal if you don't have wsl
wsl --install
wsl --update

wsl --status
```

If it says “WSL is not installed” or “Version 1,” continue with Step 2.

Step 2: Install WSL and Ubuntu 22.04

Still in PowerShell (as Administrator), run:

```
wsl --install -d Ubuntu-22.04
```

enter your username and password.

Checking WSL installation of ubuntu

```
wsl -l -v
```

this should output ubuntu 22.04.

Open vscode, and install the WSL extension from the extensions menu.

Then, press ctrl + shift + p, type wsl and choose *WSL: Connect to WSL using Distro....*

These commands will:

- Enable the WSL and Virtual Machine Platform features
- Install the latest Linux kernel
- Set WSL2 as default
- Download and install Ubuntu 22.04

After the installation completes, restart your computer.

Step 3: Launch Ubuntu

After reboot, open the Start Menu → search for Ubuntu 22.04 → click to launch it. Create a username and password when prompted.

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1. Update and Configure Ubuntu

Once Ubuntu is open, run the following commands:

```
sudo apt update && sudo apt upgrade -y
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8
```

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2. Add the ROS 2 Repository

```
sudo apt install software-properties-common -y
sudo add-apt-repository universe
sudo apt install curl -y

sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /usr/share/keyrings/ros-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) \
  signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] \
  http://packages.ros.org/ros2/ubuntu $(lsb_release -cs) main" \
  | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

3. Install ROS 2 Humble (Desktop Full)

```
sudo apt update
sudo apt install ros-humble-desktop -y
```

This installs all major ROS 2 tools (RViz, demos, tutorials, and CLI utilities).

4. Source ROS 2 Automatically

```
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

5. Install ROS 2 Development Tools

```
sudo apt install python3-colcon-common-extensions \
  python3-rosdep python3-argcomplete build-essential -y

sudo rosdep init
rosdep update
```

6. Install Gazebo Fortress (Ignition Gazebo)

```
sudo apt update
sudo apt install wget lsb-release gnupg -y

sudo wget https://packages.osrfoundation.org/gazebo.gpg \
  -O /usr/share/keyrings/pkgs-osrf-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) \
  signed-by=/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg] \
  http://packages.osrfoundation.org/gazebo/ubuntu-stable $(lsb_release -cs) main" \
  | sudo tee /etc/apt/sources.list.d/gazebo-stable.list > /dev/null

sudo apt update
sudo apt install ros-humble-ros-gz

echo 'export LIBGL_ALWAYS_SOFTWARE=1' >> ~/.bashrc && source ~/.bashrc
#verification of gazebo
ign gazebo
```

7. Install ROS–Gazebo Bridge

```
# Bridge between ROS 2 and Gazebo (Fortress or newer)
sudo apt install -y ros-humble-ros-gz-bridge
```

8. Install Important Ros2 Packages

```
#Navigation and Localization
sudo apt install ros-humble-navigation2 ros-humble-nav2-bringup -y
sudo apt install -y ros-humble-robot-localization

# SLAM and perception
sudo apt install -y ros-humble-rtabmap-ros ros-humble-imu-tools

# Example sensors (LiDAR and stereo camera)
sudo apt install -y ros-humble-rplidar-ros
```

These packages provide full path-planning, localization, and mapping functionality.

9. Install Visualization & Debugging Tools

RViz 2:

```
sudo apt install ros-humble-rviz2 -y
```

RQT Tools (for inspecting topics, transforms, etc.):

```
sudo apt install ros-humble-rqt ros-humble-rqt-common-plugins \
  ros-humble-rqt-tf-tree ros-humble-rqt-graph -y
```

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Making the ros2 workspace

```
mkdir -p ~/ros2_ws/src
cd ~/ros2_ws
colcon build
source install/setup.bash

colcon build --symlink-install
source install/setup.bash
```

Check Core ROS 2 Nodes

```
ros2 run demo_nodes_cpp talker
```

Open another terminal:

```
ros2 run demo_nodes_cpp listener
```

If you see messages, your ROS 2 setup works!

Test Gazebo Fortress

```
ign gazebo sim shapes.sdf
```

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MiniConda Environment

Follow the tutorial using this link: <https://hostman.com/tutorials/install-miniconda-on-ubuntu-22-04/>

Make sure that in the last step, you say "yes" instead of "no".

From there to use conda, execute the following commands

```
# initialize Conda
conda init

# setting the environment up
conda create -n utra_env python=3.10

# activating conda env
conda activate utra_env
```

Now we can add the install python dependencies, but before run the following command:

```
which pip
```

This should output the {Your Home Directory}/miniconda3/utra_env/bin/pip

Then execute the following commands to install the python dependencies:

```
pip3 install --user onnx onnxruntime opencv-python rospkg scikit-learn scipy
```

You're Done!

You now have:

- WSL2 and Ubuntu 22.04 fully set up
- ROS 2 Humble Hawksbill (Desktop Full)
- Gazebo Fortress Simulation Environment
- ROS-Gazebo Bridge
- Navigation 2 + Nav2 Bringup Packages
- RViz 2 and RQT Visualization Tools

You can now run, visualize, and simulate your own robots — all inside Windows.